# CHAPTER 2 KINEMATICS

**Key idea:** describe motion with $s$, $v$, $a$ and $t$; for constant acceleration use the four equations of motion.

## 1 DEFINITIONS

| | |
|---|---|
| **Displacement** | straight-line distance in a stated direction (vector) |
| **Velocity** | $v = \frac{ds}{dt}$, rate of change of displacement |
| **Acceleration** | $a = \frac{dv}{dt}$, rate of change of velocity |
| **Speed, distance** | scalars: magnitude only |

Average velocity $= \frac{\Delta s}{\Delta t}$; average speed $= \frac{\text{total distance}}{\text{total time}}$

## 2 EQUATIONS OF MOTION

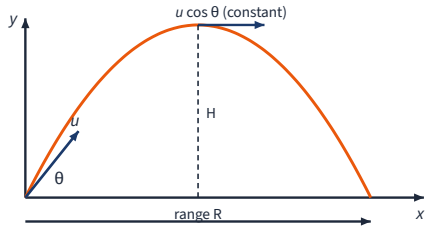
$$v = u + at \qquad s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \tfrac{1}{2}(u + v)t$$

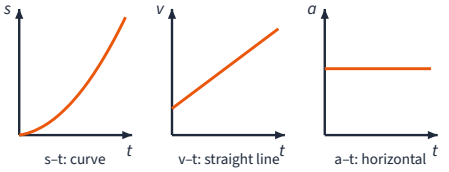
Constant acceleration in a straight line only.

## 3 PROJECTILE MOTION




| | |
|---|---|
| **Horizontal** | constant velocity $u\cos\theta$, $a_x = 0$ |
| **Vertical** | acceleration $g$ downward, $u_y = u\sin\theta$ |
| **Top** | $v_y = 0$; $v_x$ unchanged |

## 4 MOTION GRAPHS




| Graph | Gradient | Area |
|---|---|---|
| $s$–$t$ | velocity | – |
| $v$–$t$ | acceleration | displacement |
| $a$–$t$ | – | change in $v$ |

Area below the $t$-axis on a $v$–$t$ graph is negative displacement.

## 5 FREE FALL

$$a = -g = -9.81\ \text{m s}^{-2}$$

- Gravity only; air resistance neglected
- Same $g$ for all masses
- Choose **up** as positive and keep the sign
- At the highest point $v = 0$, but $a = g$ still

## 6 PROJECTILE FORMULAS

$$T = \frac{2u\sin\theta}{g}$$

$$H = \frac{u^2\sin^2\theta}{2g}$$

$$R = \frac{u^2\sin 2\theta}{g}$$

- $R_{\max}$ at $\theta = 45^\circ$; $\theta$ and $90^\circ - \theta$ give the same range
- Path is a **parabola**: $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

## 7 HORIZONTAL LAUNCH FROM HEIGHT H

$$t = \sqrt{\frac{2h}{g}} \qquad x = ut \qquad v = \sqrt{u^2 + (gt)^2}$$

- Time of fall depends on $h$ only, not on $u$
- Angle below horizontal: $\tan\phi = \frac{gt}{u}$

## 8 COMMON SECTION A TRAPS

- Mixing signs: if up is +, then $g$ is −9.81 m s$^{-2}$.
- Using $R = u^2\sin 2\theta/g$ when landing at a **different** height.
- Saying acceleration is zero at the top of the path.
- Treating distance as displacement on a return journey.

## EXAM TIPS

- ✓List $s, u, v, a, t$; pick the equation without the unknown you do not need.
- ✓Split projectile motion into horizontal and vertical parts.
- ✓Time links the two directions.
- ✓State the direction of every vector answer.